

RAG PROJECT

Ontology-Aware Clinical RAG System for Safe and Evidence-Grounded Medical Information Retrieval

The 3 Core Parts Explained

1. Clinical RAG (Retrieval-Augmented Generation):

- **What it means:** The AI doesn't just rely on its memory. Before answering, it searches through trusted medical databases and peer-reviewed journals to pull up exact evidence.
- **Analogy:** It's like an open-book exam where the AI checks the latest medical textbooks before giving an answer.

2. Ontology-Aware:

- **What it means:** The AI understands how medical concepts strictly relate to one another (using standard medical dictionaries like SNOMED CT or UMLS). It knows that *Hypertension* is the exact same thing as *High Blood Pressure*, and it understands that *Medication A treats Condition B*, but *should never be prescribed alongside Medication C*.
- **Analogy:** It acts like a built-in "medical logic checker" that prevents the AI from confusing terms or recommending dangerous combinations.

3. Safe & Evidence-Grounded:

- **What it means:** The final response is directly tied to verified medical sources, minimizing the chance of hallucinations (made-up facts) and prioritizing patient safety.

The Core Concept

- **AIOps (AI for IT Operations):** Using AI to monitor servers, apps, and cloud networks, catch problems automatically, and keep everything running smoothly.
 - **Multi-Agent System:** Instead of relying on one single AI model, it uses a team of specialized AI "agents" working together—like having one agent analyze logs, another draft a fix, and another verify safety.
 - **Autonomous Incident Remediation:** The system doesn't just send an alert to an engineer; it actively fixes the server or application issue on its own.
-

The 3 Key Features Explained

1. Confidence-Aware Hybrid Routing:

- **What it means:** The system measures how sure it is about a fix. If it's high-confidence, it handles the task using cheap/fast models or automated scripts. If it's low-confidence or complex, it routes the task to a smarter AI model or escalates it to a human engineer.
- **Analogy:** Like a triage nurse who treats minor cuts directly but immediately sends complex cases to a specialist surgeon.

2. Safety-Governed Execution:

- **What it means:** Built-in safety protocols check every command before it runs. If the AI proposes a risky fix (like deleting a crucial database), guardrails stop it from executing without approval.
- **Analogy:** Putting a "double-check" safety lock on the emergency stop button so the AI can't accidentally pull the plug on the wrong server.

3. Reflection-Based Recovery:

- **What it means:** If the AI's initial fix fails, it doesn't give up or repeat the same mistake. It "reflects" on the error log, learns what went wrong, and adjusts its strategy to try a better solution.
- **Analogy:** A mechanic trying a fix, checking if the engine starts, realizing why it didn't work, and adjusting their approach based on the new symptom.

In Short

MARIS is a smart, team-based AI system that automatically repairs IT server crashes. It knows when to handle a fix automatically versus when to ask for help, double-checks commands for safety, and learns from failed attempts on the fly.